Predicting Footballer Market Values

Case Study for Predictive Modelling

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Objective

Can detailed player skills explain market value — and does the relationship change by playing role?

Build and evaluate models that estimate FIFA 23 player market value using detailed skills, age, playing role and potential.

- Can the skill rating parameters be summarized into a smaller set of useful dimensions?
- Do those dimensions reveal broad groups of players with similar profiles?
- Do role-specific models predict market value better than one model for everyone?

Data Overview & Preprocessing

Dataset Description

Total Players
161,583

Variables
110

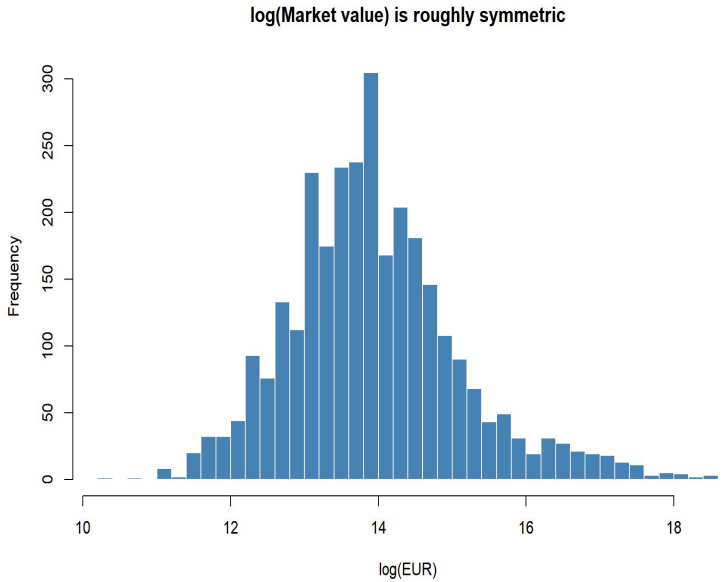
Skill Related Variables
29

Target Year
FIFA 23

Key Preprocessing Steps

- **Filtering:** Focused on outfield players (GK excluded) with non-zero market value.
- **Log Transformation:** Market value is extremely right-skewed; log(value) was used to achieve symmetry for regression.
- **Sampling:** 3,000 players for modeling, 13,396 for hold-out validation.
- **Feature Engineering:** Grouped players into 4 functional roles (DEF, DM, MID, FWD).

Market Value Distribution



Note: The raw market value follows a power-law distribution, necessitating logarithmic scaling for linear modeling.

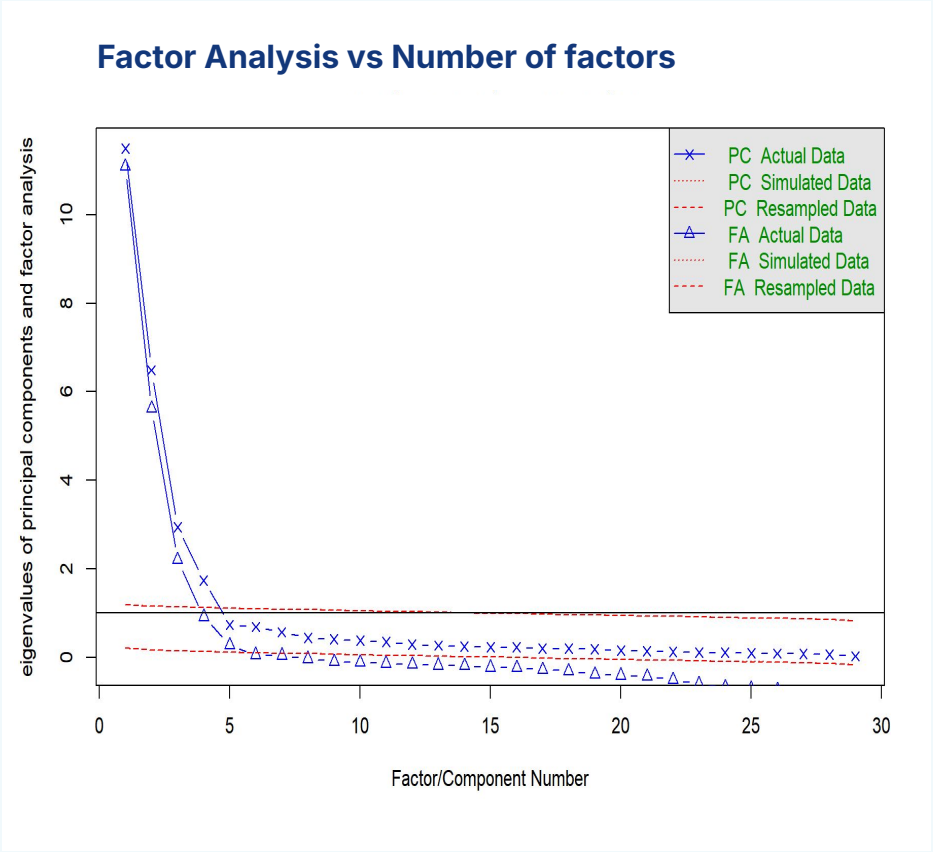
PCA - choosing number of dimensions

Principal Component Analysis (PCA) combines correlated ratings into a smaller set of weighted scores. It builds new variables, each a weighted combination of the 29 skills. Parallel analysis compares the scree plot with simulated and resampled data.

Parallel analysis suggested 4 components for PCA (and 5 factors for common-factor analysis). We retain 4 here because we use PCA.

We are able to retain **78.1%** of variance from these 4 new reduced parameters using varimax rotation

Each component is a weighted combination of the original skills—not a pre-defined FIFA category.



Interpreting the four components

PCA does not name its components. We label each one from its highest loadings.

Every player receives a continuous score on all four dimensions.

Technical (36.2%) : Long shots, vision, ball control, shot power.
Defending (19.5%) : Standing tackle, sliding tackle, interceptions, marking
Pace (12.3%) : Acceleration, sprint speed, agility, balance.
Physical (10.1%) : Heading accuracy, strength, jumping.

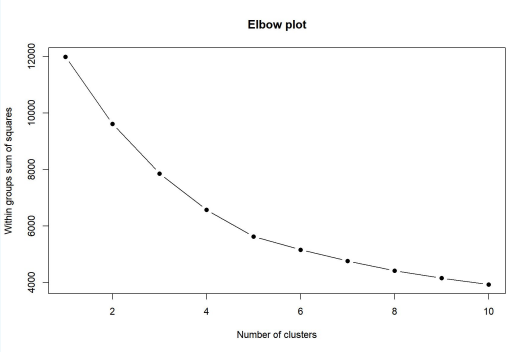
29 vs 4 param correlations

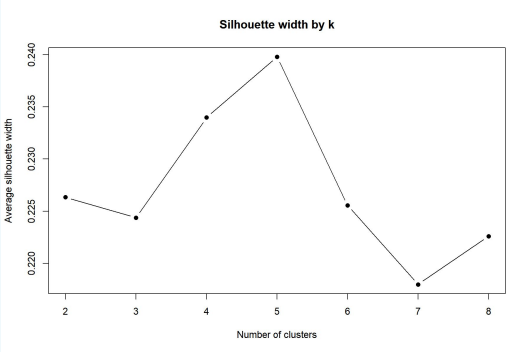


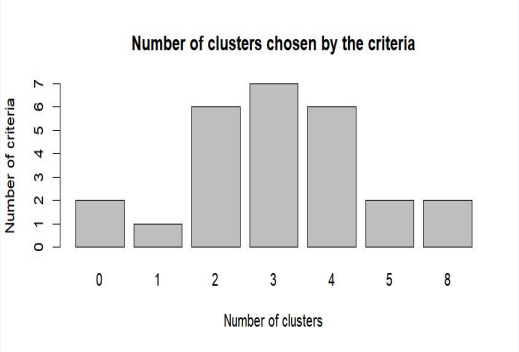
K-means clustering

PCA : What are the main dimensions of footballing ability?

K-means : Given those dimensions, which players have similar ability profiles?







within-cluster sum of squares (WSS) vs k

silhouette scores vs k

nbclust voting vs k

WSS : which measures how far players are from their assigned cluster centres.

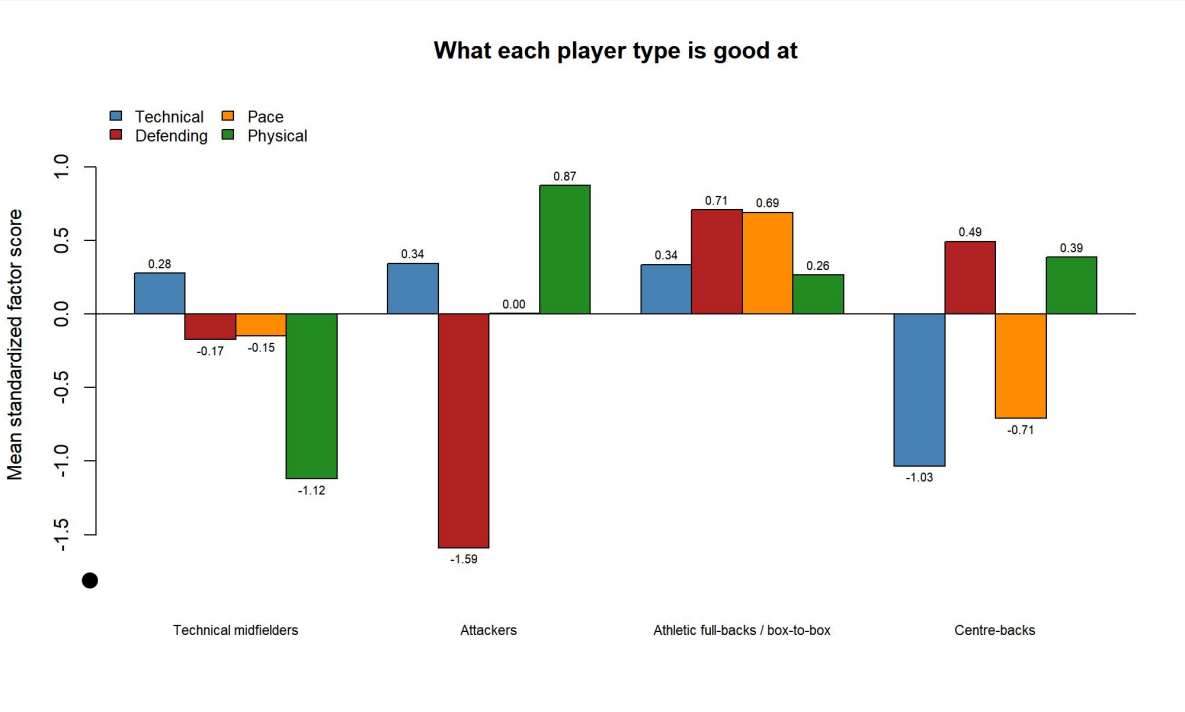
Silhouette : checks whether players are well matched to their own cluster and reasonably separated from neighbouring clusters

NbClust : list of 28 criterions choosing their best suitable k vals

Player types: cluster profiles

Positions were not used to build the clusters, yet cluster purity against position is 69% (37% baseline):

- 73% of forwards are Attackers
- 53% of defenders are Centre-backs



Technical midfielders
n = 866 · overall 63 · age 23 · €0.78M

Attackers
n = 526 · overall 66 · age 25 · €1.10M

Athletic full-backs / box-to-box
n = 907 · overall 69 · age 26 · €1.60M

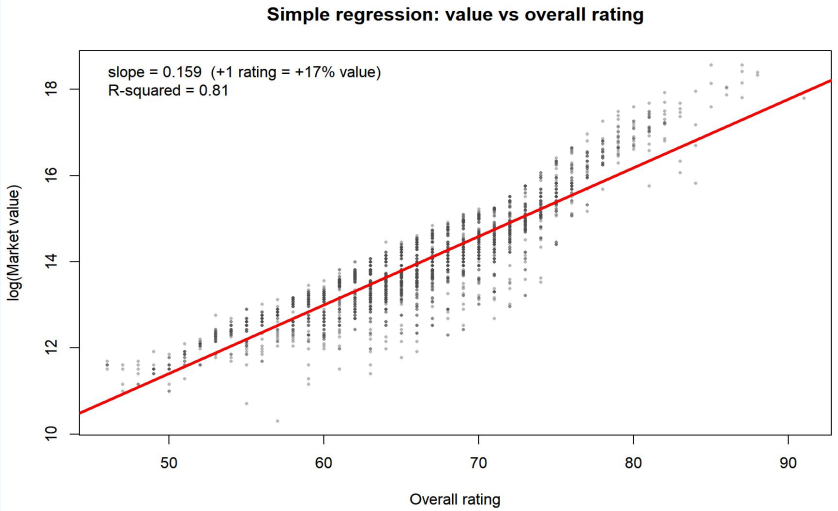
Centre-backs
n = 701 · overall 64 · age 25 · €0.72M

Regression

Model Hierarchy

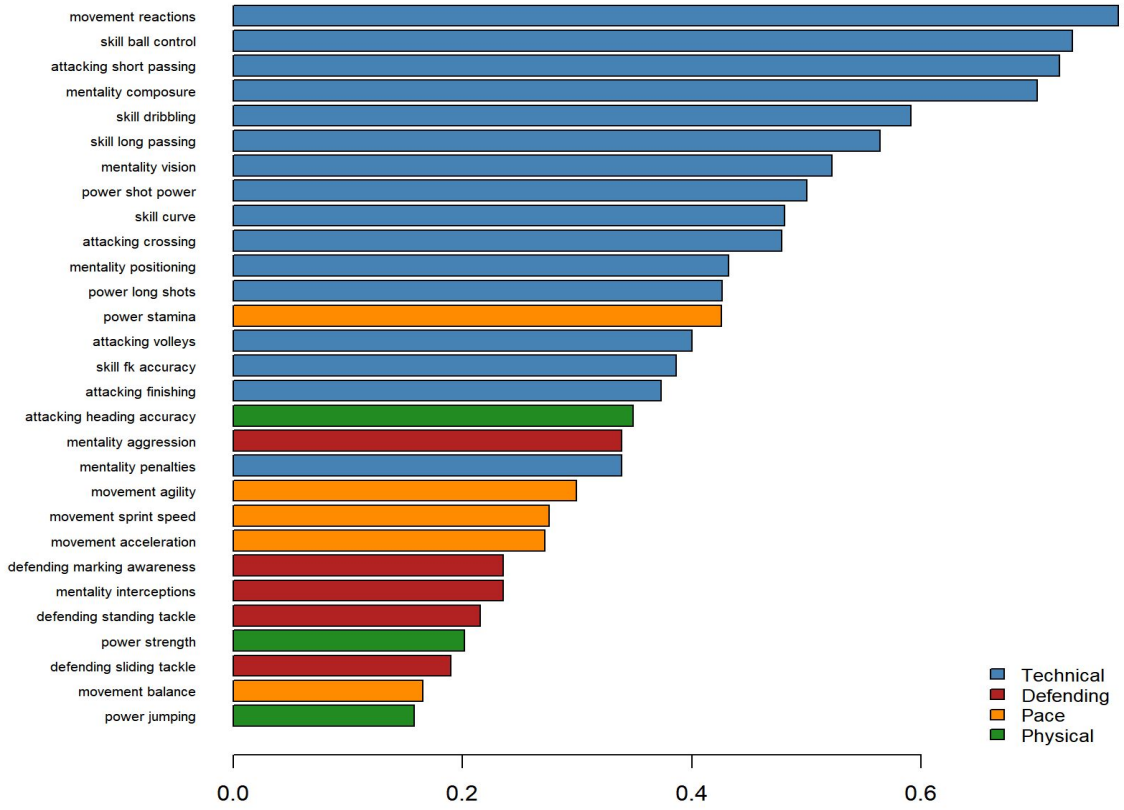
Model	Description
M1	Baseline using only "Overall" rating.
M2/M4	Pooled model with all 29 skills + Age + Position.
M3	PCA-based model using 4 Skill Dimensions.
M6	**Per-Role Models:** Separate coefficients for DEF, DM, MID, FWD.
M5/M7	M4/M6+ Player Potential.

Ratings vs market value prediction

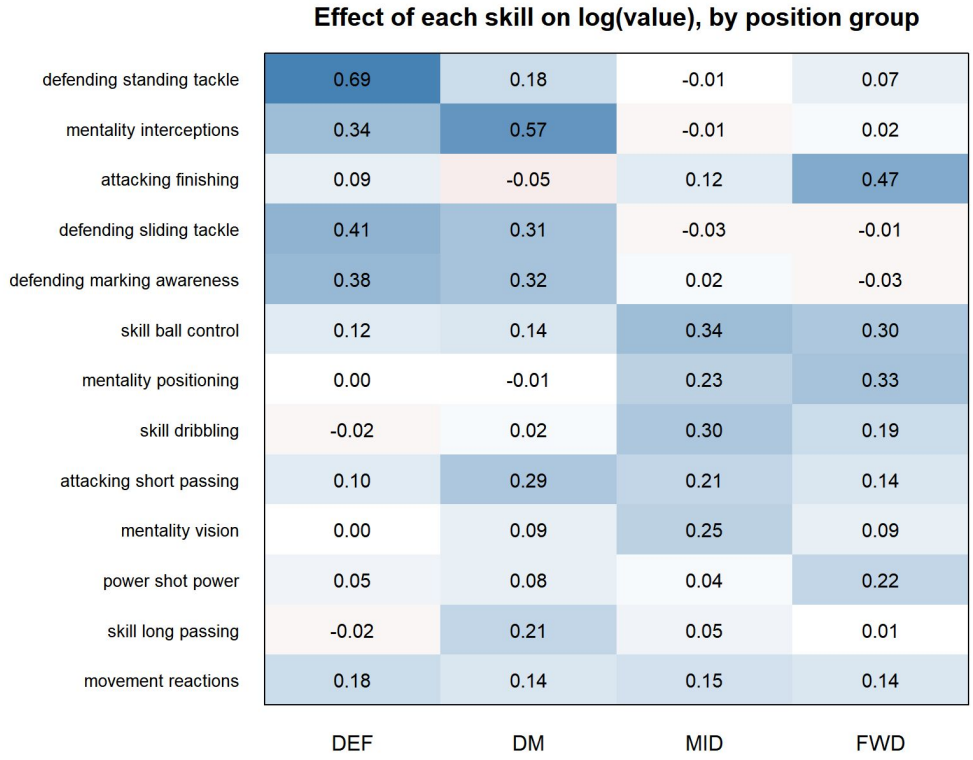


R^2 = 0.81 from overall rating alone: each extra point adds about 17% to value. Adding age, position and the 29 skills lifts R^2 to 0.86.

Which skills correlate with market value overall?



The skills associated with value differ by role



DEF : Standing tackle (0.69), sliding tackle, marking.

DM : Interceptions (0.57), marking, sliding tackle.

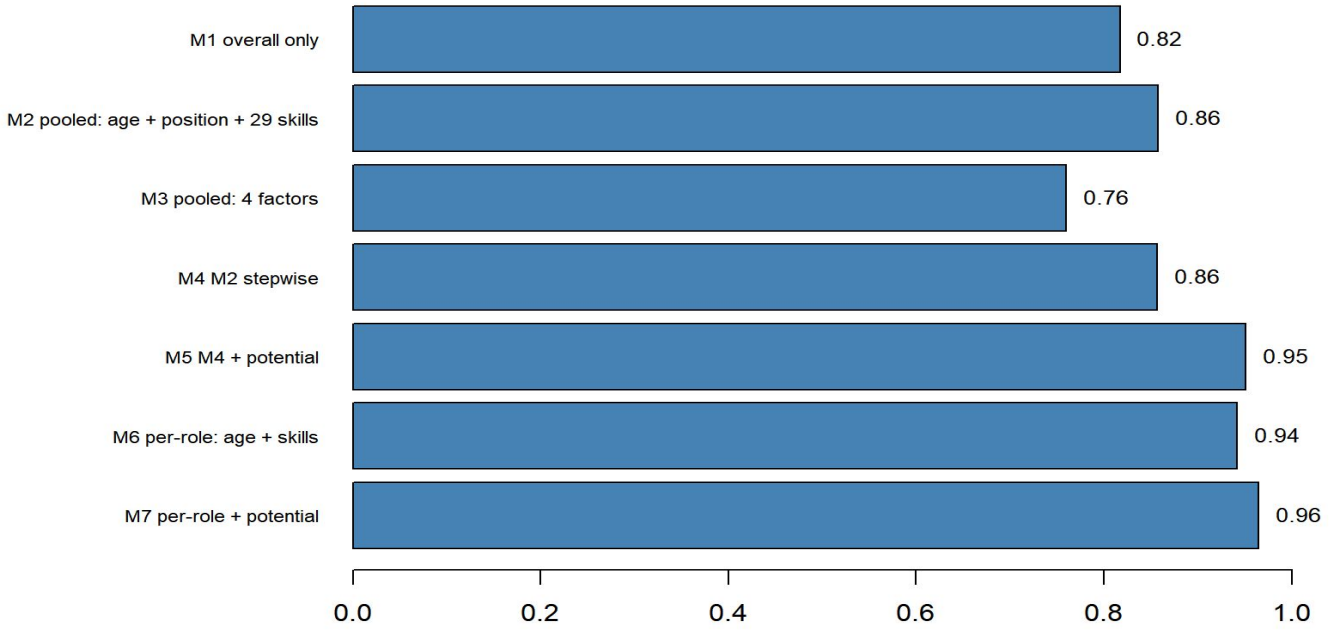
MID : Ball control (0.34), dribbling, vision.

FWD : Finishing (0.47), positioning, ball control.

In the pooled model, positioning has correlation +0.43 but coefficient −0.14. Per position it is +0.23 (MID) and +0.33 (FWD).

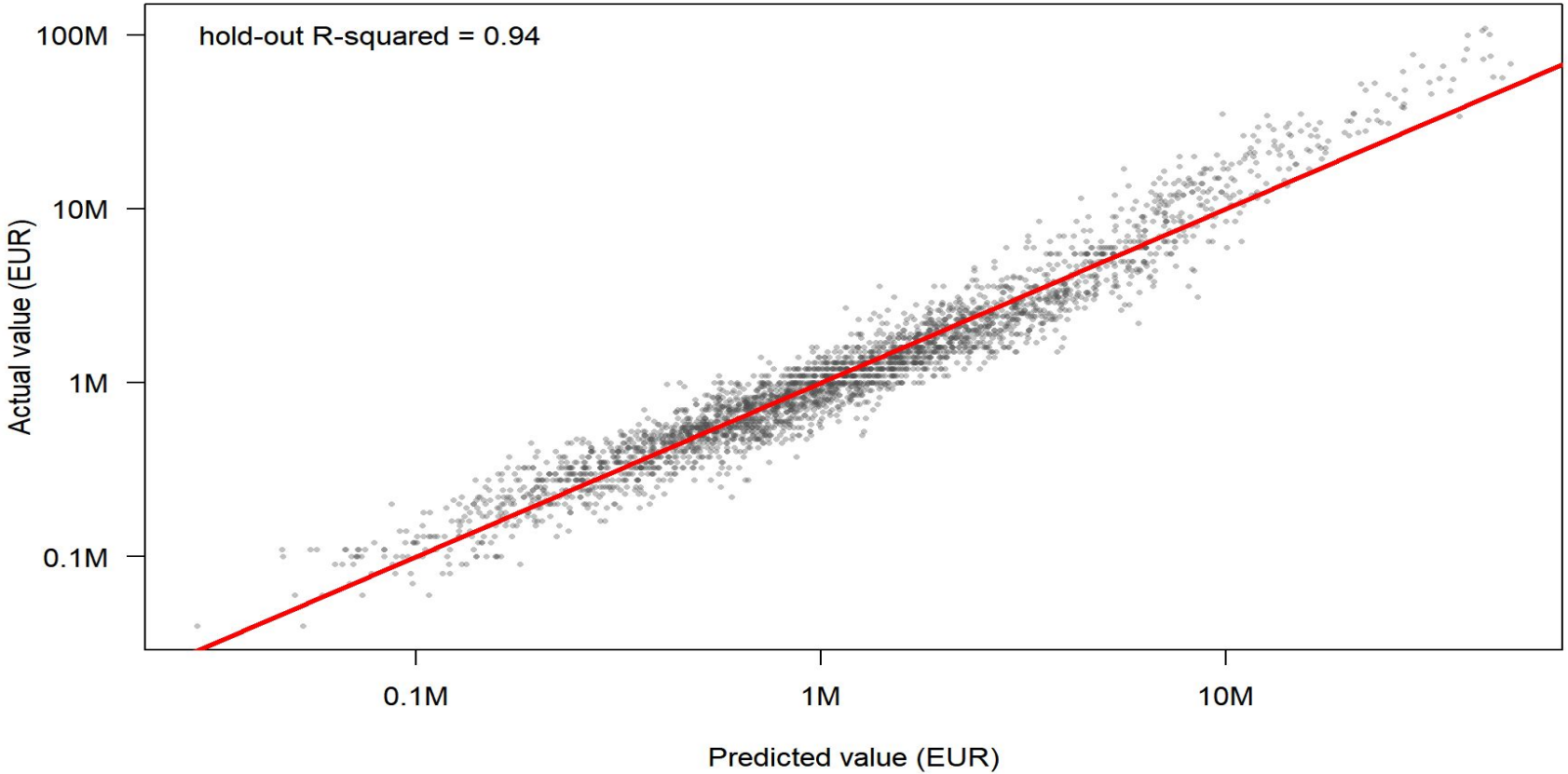
Model Performance Comparison

BASELINE R^2 (M1)	BEST SKILLS-ONLY R^2 (M6)	FINAL MODEL R^2 (M7)
0.817	**0.942**	**0.965**



R-squared measured on a 13,000+ player hold-out set. The "Per-Role" approach (M6/M7) significantly outperforms the "Pooled" approach (M2-M4).

Predictive Accuracy on Hold-out Data



Log-log scale plot of Predicted vs. Actual market values (EUR) for 13,396 unseen players. Red line indicates perfect prediction (y=x).

Market Outlier Analysis: Premiums & Discounts

Market Premiums (Overvalued)

Players valued significantly higher than their skills justify (Brand/Hype premium).

Player	Actual	Expected	Gap
A. Tchouaméni	€61.5M	€20.2M	**+€41.3M**
K. Koulibaly	€54.5M	€22.4M	**+€32.1M**
I. Konaté	€48.0M	€20.1M	**+€27.9M**
Pablo Sarabia	€25.0M	€9.8M	**+€15.2M**
Manu Trigueros	€17.0M	€5.1M	**+€11.9M**

Market Discounts (Undervalued)

Players with elite skill profiles but low market valuation (Scouting Opportunities).

Player	Actual	Expected	Gap
Lim Min Hyuk	€2.5M	€8.4M	**-€5.9M**
J. Riedewald	€4.8M	€10.2M	**-€5.4M**
D. Mertens	€12.5M	€18.1M	**-€5.6M**
S. Ajayi	€3.2M	€7.8M	**-€4.6M**
V. Grifo	€28.5M	€32.1M	**-€3.6M**

Diagnostic Insight: Residual analysis identifies players like *Tchouaméni* as having a "Real Madrid premium" where market value exceeds statistical contribution. Conversely, *Riedewald* and *Mertens* represent statistical "bargains" whose skills are undervalued by the market.

Conclusions & Final Insights

Role-Specific Dynamics

Market valuation is not universal; a per-role modeling approach is essential to capture the context-dependent value of player skills.

Predictive Power

Our final model (M7) achieved a hold-out R-squared of 0.965, accurately predicting the value of 13,000+ unseen players.

Scouting Utility

Residual analysis effectively identifies market inefficiencies, spotting "branding premiums" and "statistical bargains."

"Predictive modeling demonstrates that professional football market values are largely driven by a structured interaction between specific skill sets and positional requirements, supplemented by potential and age."

Final R^2: 0.965

Dataset: FIFA 23

Thank You!